

Semantic core

What is true, under what conditions?

Stable claim

With negligible air resistance and constant g , vertical velocity changes uniformly: $v_y = u_y + a_y t$; with upward positive, $a_y = -g$.

Apex boundary

At the apex $v_y = 0$ but $a_y = -g$; horizontal velocity may remain nonzero.

Authority boundary

Core1 does not choose pedagogy, infer mastery, or authorize transfer.

Representation invariants

Same meaning across forms.

Story

Gravity changes v_y by the same signed amount each second.

Table/graph

Equal time steps give equal Δv_y ; the v_y -t graph is a straight line whose slope is a_y .

Misconceptions

Downward acceleration does not mean downward motion; a v_y -t graph is not the trajectory.

Core1 handoff

Stable semantics only.

Downstream

Core1A teaches; Core1B gathers learner evidence; Core2 describes challenge structure; Core2A authorizes transfer; Core2B runs transfer.